INVENTIX - EVENT MANAGEMENT SYSTEM

System Requirements Specification & Architectural Blueprint

Project Academic Context: Bachelor 3 Software Engineering

Institution: IAI Cameroon / CET Paul Biya

Core Framework: Django (Python) + Relational SQL

Target Deployment: Cameroon Event Ecosystem

1. PROBLEM STATEMENT & CONTEXT

In Cameroon, organizing events across academic institutions, professional hubs, and social spaces remains heavily dependent on fragmented, manual workflows. Event organizers rely on unstructured WhatsApp broadcasts, physical sign-up sheets, cash transactions, and printed clipboard guest lists. This operational fragmentation causes four severe structural bottlenecks:

- **Capacity Mismanagement:** Lack of real-time seat tracking causes severe overcrowding in university halls or catering shortages at private events.
- **Ticketing Fraud & Gate-Crashing:** Paper passes and manual messaging receipts are easily replicated, leading to unauthorized entry.
- **Check-in Bottlenecks:** Manual name-searching at venue entrances creates long queues under adverse weather conditions.
- **Reconciliation Inefficiencies:** Manual cash and mobile money tracking leads to post-event financial discrepancies.

2. LOCAL EVENT TAXONOMY & CORE PROBLEMS

Category	Typical Events	Existing Methods	Core Operational Problem
Academic & Professional	Conferences, Seminars, Symposia	Google Forms, Paper Sign-ins	Uncapped registration, venue overcrowding, slow entry.
Entertainment & Cultural	Concerts, Festivals, Galas	Paper Tickets, MoMo Screenshots	High counterfeit risk, gate-crashing, cash discrepancies.
Recreational & Sports	Football, Basketball Matches	Stadium Booth Cash Sales	Long queues, no pre-registration, security hazards.
Private Milestones	Marriages, Birthday Parties	Paper Cards, WhatsApp RSVP	Inaccurate catering headcounts, uninvited gatecrashers.

3. DIGITAL WORKFLOW FOR PERSONAL EVENTS (MARRIAGES & BIRTHDAYS)

Phase 1: Setup & Seating Limit
The host registers the event, sets venue coordinates, and defines a strict maximum seating/catering capacity in the database.

Phase 2: Digital Pass Dispatch
Inventix generates unique cryptographic invitation tokens and emails/dispatches QR pass links to specific guest contact lists.

Phase 3: Real-time RSVP Locking
Guests confirm or decline attendance via personalized pages. Confirmed RSVPs instantly reserve database seats until capacity is reached.

Phase 4: Instant Door Verification
Ushers scan guest QR codes at the entrance using the Inventix web view, verifying legitimacy in milliseconds to eliminate gatecrashers.

4. SYSTEM ACTORS

- **Guest / Visitor:** Unauthenticated user who browses public listings, searches categories, and creates an account.
- **Attendee / Participant:** Authenticated user who books tickets, manages personal RSVPs, and accesses digital QR passes.
- **Event Organizer:** Authorized user who creates events, sets capacities, tracks attendance, and validates entry at the door.
- **System Administrator:** High-privilege user responsible for user roles, event moderation, and database administration.

5. CONCRETE SYSTEM USE CASES

UC-1: Public Event Discovery
Guests filter upcoming events by category, date, and city (e.g., Yaoundé, Douala), viewing ticket availability and venue details.

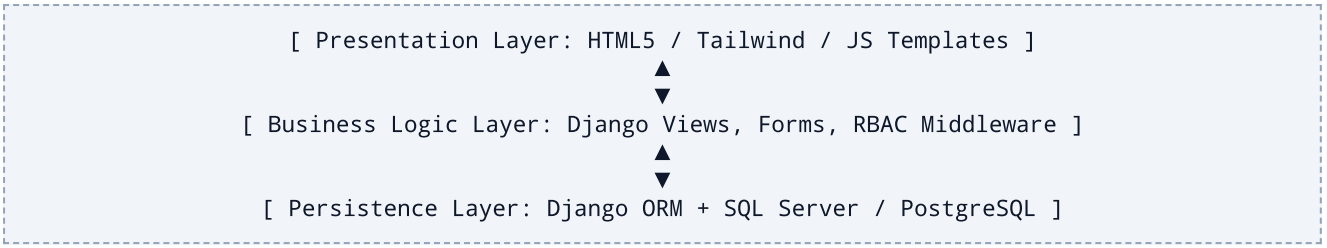
UC-2: Capacity-Controlled Booking
Attendees select ticket counts; the backend uses SQL transactions to ensure seats are allocated without overbooking.

UC-3: Event Creation & Dashboard
Organizers define venue boundaries, ticket tiers, pricing, and view real-time registration/revenue analytics.

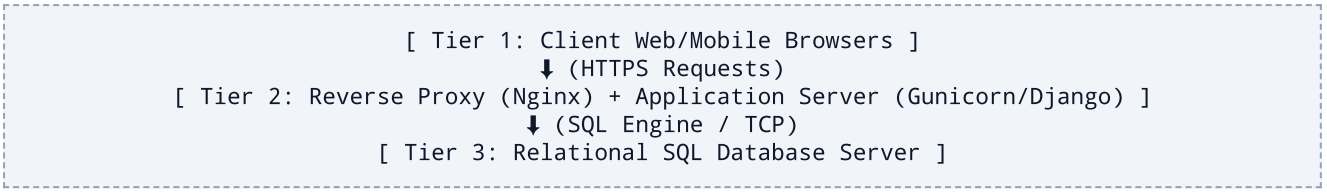
UC-4: Door Check-In Validation
Organizers scan attendee QR codes. The system checks token validity and marks attendance instantly.

6. SYSTEM ARCHITECTURES

Logical Architecture (Django MVT):



Physical Architecture (3-Tier Model):



7. CURRICULUM-ALIGNED TECH STACK

Layer	Selected Technology	Curriculum Alignment
Backend Framework	Django (Python 3.x)	Advanced Object Oriented Programming
Database Engine	SQL Server / PostgreSQL	Oracle & SQL-Server DB Administration
Data Access	Django ORM	Advanced Python Programming
Frontend UI	HTML5, CSS3, JavaScript	Web Programming Tools
Security & Auth	Django Auth & RBAC	Database Security

8. CORE APPLICATION APIS

- **Auth API:** Handles registration, login JWT/session tokens, and role authorization.
- **Events API:** Provides endpoints for event CRUD, category filtering, and real-time seat queries.
- **Booking API:** Executes ticket reservations, generates unique QR tokens, and releases cancelled seats.
- **External Integrations:** QR generation library, SMTP Email dispatcher, and Mobile Money webhooks.